L2 - Heaps

Special class

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→ K^{th} smallest element

$K = 3$

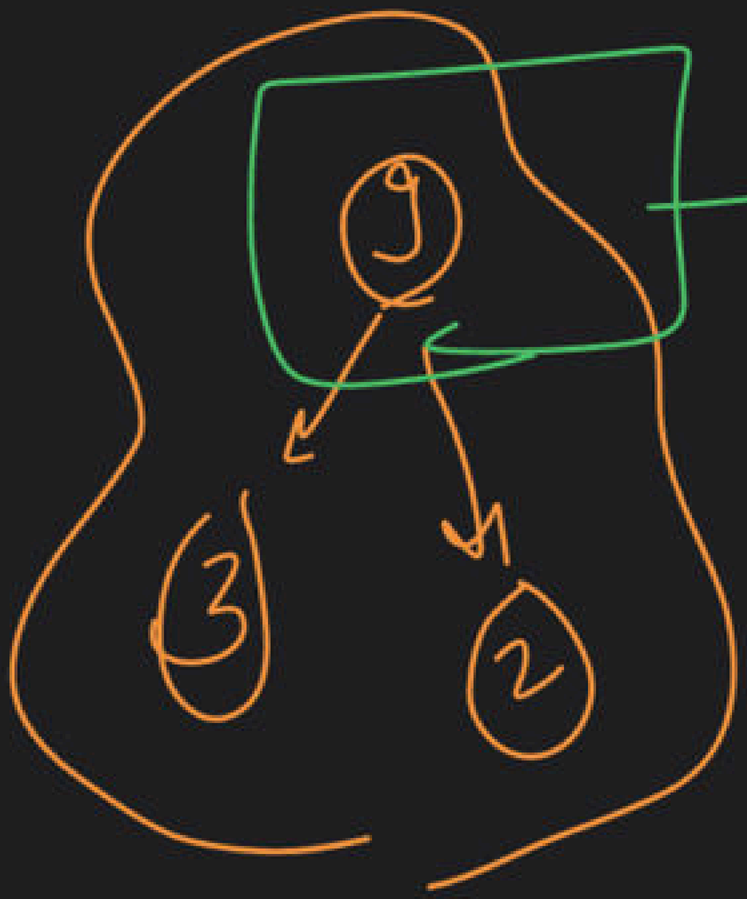
ignore ignore ignore

10	3	2	50	14	17	9
0	1	2	3	4	5	6

K - size → Man-Heap

chotte elements in heap

initially



$S.C \to O(K)$

K^{th} smallet element

$T.C \to n \log K$

→ Min Cost Ropes → Greedy Algo { sort, min/max, heap, set }

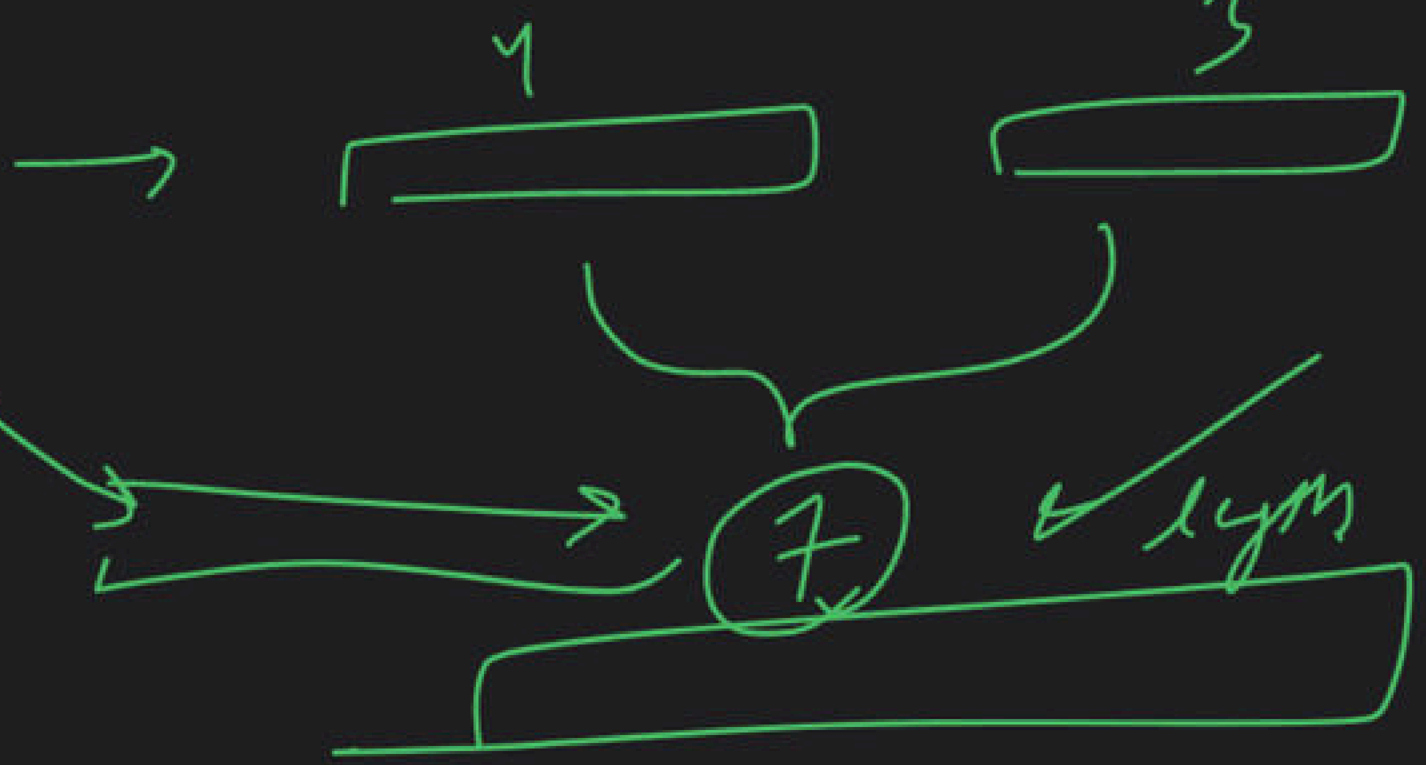
Approach

{ ~~...~~, ~~...~~, ~~...~~, 15 }

2+3 → 5
4+5 → 9
6+9 → 15

5 + 9 + 15 = 29 ans

cost →



minimum total cost

Total cost = 0 + 5 + 9 + 15

element1 = 2
pop
element2 = 3
pop
→ 5

Element 1 = 4
pop
Element 2 = 5
pop
→ 9

element1 = 6
pop
element2 = 9
pop
→ 15

min
Heap

15

size = 1
Return
Total cost
29
ans

4, 3, 2, 6